

HAT-P-3b

R-1335

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-3_200519 · created 2026-07-31T21:00:33+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458988.8333 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1119 +/- 0.0083
Transit depth (Rp/Rs)^2	1.25 +/- 0.19 [%]
Orbital Inclination (inc)	89.64 +/- 0.64
Airmass coefficient 1 (a1)	0.9815 +/- 0.006
Airmass coefficient 2 (a2)	0.0137 +/- 0.0062
Scatter in the residuals of the lightcurve fit is	0.61 %
Best Comparison Star	#3 - [540, 170]
Optimal Aperture	4.55
Optimal Annulus	11.95
Transit Duration (day)	0.0982 +/- 0.0013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1119 ± 0.0083 vs 0.1063 (deviation 0.61σ)
Duration fit vs published	0.0982 d vs 0.0867 d (deviation 7.99σ)
Mid-time O–C	6.7 min
Depth SNR	12.1
Residual scatter	0.61 %

Light curve

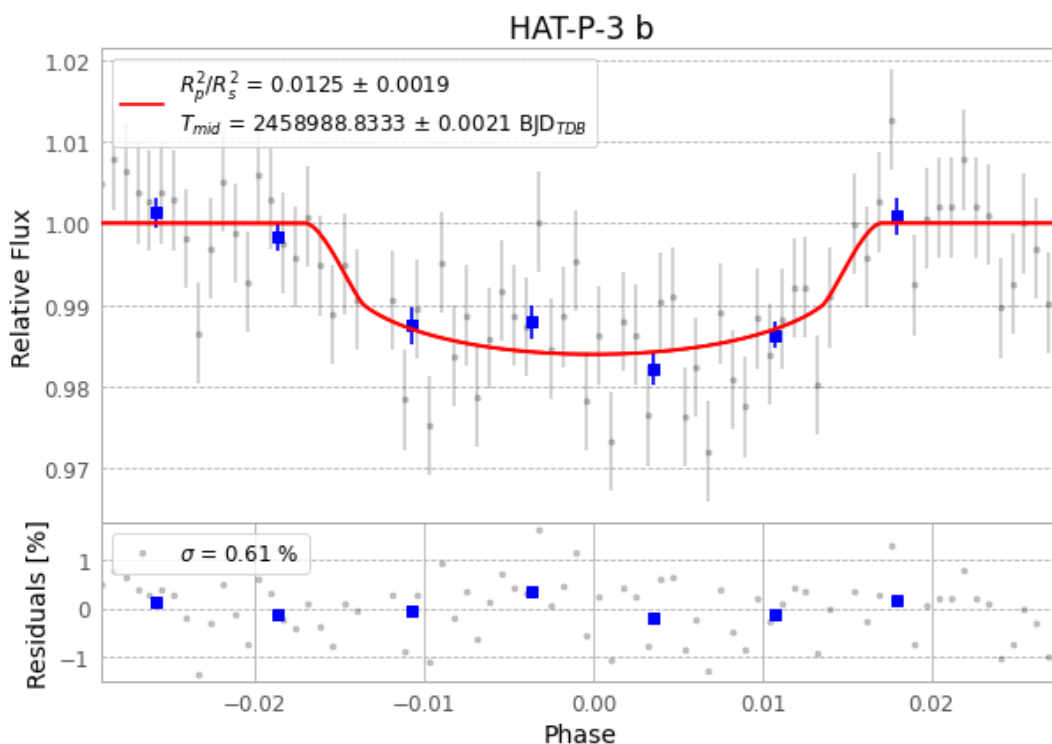


Figure 1 — FinalLightCurve_HAT-P-3 b_2020-05-18.png (SHA-256 34903e6bbcc53939...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [458, 249], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c9b6502c9e99a13c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

78 FITS frames (51 MB), HATP-3200519055412.FITS → HATP-3200519095112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-3-200519.log (65e6221abc99...), FinalLightCurve_HAT-P-3 b_2020-05-18.png (34903e6bbcc5...), FinalLightCurve_HAT-P-3 b_2020-05-18.csv (b7a398c827b0...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 21:00Z.